

1 hour on examples

Problem 3 (GRTH5)

- a) We double count, since each edge adds 1 to the two vertices it connects,

$$\sum_v \deg(v) = \sum_e 2$$

- b) We once again double count. Each edge adds one to the indegree sum and one to the outdegree sum.
c) We use induction to prove this.

Base Case: $n = 1$, $0 = 0$ is obviously true.

Inductive Step: $n \rightarrow n + 1$,

For any tournament with $n + 1$ vertices, we may look at the tournament with n vertices that we get when we remove an arbitrary vertex as well as all the edges connected to it from the graph.

The resulting graph currently satisfies,

$$\sum_v (\text{indeg}(v))^2 = \sum_v (\text{outdeg}(v))^2$$

When we add on a new vertex m , we want,

$$\begin{aligned} \sum_v (\text{indeg}(v))^2 &= \sum_v (\text{outdeg}(v))^2 \\ \Leftrightarrow \sum_{v \neq m} (\text{indeg}(v))^2 - (\text{outdeg}(v))^2 &= \text{outdeg}(m)^2 - \text{indeg}(m)^2 \end{aligned}$$

Since $\sum_{v \neq m} (\text{indeg}(v))^2 - (\text{outdeg}(v))^2$ was previously 0, we just prove the change in this value is $\text{outdeg}(m)^2 - \text{indeg}(m)^2$.

Let's say m has k edges pointing towards vertices which previously had indegree $\{d_1, d_2, \dots, d_k\}$.

Therefore, $\sum_{v \neq m} (\text{indeg}(v))^2$ will get incremented by $\sum (d_i + 1)^2 - d_i^2 = \sum 2d_i + 1$.

Let the indegrees of the remaining vertices be $\{d_{k+1}, \dots, d_n\}$. The outdegrees of these vertices would hence be $\{(n - 1) - d_{k+1}, \dots, (n - 1) - d_n\}$. Therefore, $\sum_{v \neq m} (\text{outdeg}(v))^2$ will be incremented by $\sum (n - d_{k+1})^2 - (n - 1 - d_{k+1})^2 = \sum (2n - 1) - 2d_i$.

Therefore, we're just proving,

$$\begin{aligned} \sum_{i=1}^k 2d_i + 1 - \sum_{i=k+1}^n (2n - 1) - 2d_i &= k^2 - (n - k)^2 \\ \Leftrightarrow \left(\sum 2d_i \right) + k - (2n - 1)(n - k) &= 2nk - n^2 \\ \Leftrightarrow (n^2 - n) + k - (2n^2 - n + 2nk + k) &= 2nk - n^2 \end{aligned}$$

Problem 5 (CGMO 2012/6)

This is equivalent to whether we may color the edges of a graph with n vertices red or blue such that we don't have any cycles of more than 3 edges with the same color. We call such cycles "illegal".

We claim the maximum is $n = 4$.

Let the vertices be conveniently numbered from 1 to n .

$n = 3$ is achieved when we color $1 \leftrightarrow 2$ red, $2 \leftrightarrow 3$ red, and $1 \leftrightarrow 3$ blue.

$n = 4$ is achieved when we color $1 \leftrightarrow 3$ red, $1 \leftrightarrow 2$ red, and $2 \leftrightarrow 4$ red. All other edges may be colored blue.

We now prove that $n = 5$ is impossible. For any greater n , there'd be an illegal cycle in the complete graph formed by any subset of 5 vertices of the graph. For $n = 5$, if there are 5 edges of the same color, then there must be a cycle. By Pigeonhole, we are done.

5 clubs; 2.5 hours

Problem 7 (SICKNESS)

- a) We can let the students be on a graph G with friendships as edges and each student as a vertex. The students who start out as infected can be called patient zeros. Note that any student who are not in the same connected component as any patient zeros will never be infected, so we can ignore the connected components they're a part of.

We label each vertex by the minimum distance (in edges) to a patient zero [quick question are we allowed to state it like this, it makes tremendous sense to me since it's just BFS, but it is obvious that this is do-able right. We claim that on Day n , the set of sick vertices is exactly the vertices labeled $n - 1$.

Base case: Day 1, only patient zeros are sick.

Inductive step: From day k to day $k + 1$

On day k for $k \geq 2$. By the induction hypothesis, on day $k - 1$, the students that were previously sick and now immune are the vertices labeled $k - 2$. The students that are currently spreading the sickness are the ones labeled $k - 1$. Since each vertex labeled $k - 1$ can only be adjacent to vertices labeled $k - 2$ and k , on the next day, day $k + 1$, only the vertices labeled k can be sick.

Conclusion: By mathematical induction we're done.

- b) Have 3 vertices labeled A, B, C forming a complete graph. Vaccinate A, have B start sick, and the virus will rotate around.

Problem 10 (CZHGT316)

There's a bijection between every isomorphically distinct graph of 9 vertices where every vertex has degree 6 and every isomorphically distinct graph of 9 vertices where every vertex has degree 6 and every isomorphically distinct graph with 9 vertices where every vertex has degree 2.

This is because for a given graph, we can let function f be taking the complement by drawing all the edges between two distinct points and then erasing the edges we started with. Since $f(f(g)) = g$ for all graphs g , f is bijective.

Since the degree of all the edges is 2, then the graph must consist of k connected components, each of which must be exactly one cycle (since graph is simple, each cycle must be length at least 3) [don't know if I have to prove this; also don't know what this is called].

We can casework on number of connected components, to get our answer,

$$= 1 + 2 + 1 = 4$$

15 clubs; 3.5 hours

Problem 12 (19PUMACFA1)

We consider each connected component independently. We prove that each connected component will eventually become a tree such that there are no cycles.

For a connected component with N vertices and M edges ($N \leq M - 1$ for it to be connected), first note that after each operation, each vertex and the new vertex are still connected. During each operation, also note that N increases by one while M doesn't change.

Therefore, once we have $N = M - 1$, the connected component will be a tree and we're done.

Problem 13 (23CANWINTERCAMP)

FTSOC, let a connected graph be split.

Note that each color is connected to each vertex exactly once means for each color, we may form pairs of vertices to connect with the edge of that color. Therefore, our connected graph must have started with an even number of vertices.

Note that if one of the connected components X we get after the division has an even number of vertices, then for any color, the edge we took out may not lead from X to another connected component since $X - 1$ is odd (!).

Therefore, all the connected components after the removal of the different colors must have an odd number of vertices. However, now the color that was not removed may no longer fit into all these odd, connected components.

25 clubs; 5 hours

Problem 14 (16TMD8) Unique solution, I think! Idk if it works though :/

We start with coloring all the vertices purple. We start by choosing any vertex to color green. We then color all the vertices adjacent to it blue.

We repeat the following steps until there are no more purple vertices: Find a purple vertex that is adjacent to a blue vertex and no green vertices. Color all the vertices adjacent to it blue.

Obviously, there cannot be any green vertices adjacent to this purple vertex otherwise the vertex wouldn't have been purple. Therefore, if a purple vertex exists, there exists a purple vertex adjacent exclusively to blue or purple vertices. Since the number of purple vertices is decreasing, we'll eventually be done.

We can now color all the edges between green and blue edges black. Since, using this process, all the green vertices we add are connected to a previously added green vertex via one blue vertex, all the green vertices are connected to each other. We also have all the blue vertices are connected to a green vertex, so the black edges form a connected graph.

Finally, we connect the remaining edges white. Since we don't have any green-green edges, white edges have at least one blue endpoint and we're done.

Problem 16 (Balkan 2013/4)

We can do each connected component of G independently. We hence just prove this for each connected component, starting with looking at each chord-less cycle in the graph.

Claim: There is a vertex of degree ≤ 2 .

Proof: FTSOC if not, we look at one of the paths of maximal chord-less length $x_1 \rightarrow x_2, \dots, \rightarrow x_k$. Since the degree is ≥ 3 , x_1 must be adjacent to two other vertices u, v . Since the path is maximal, u and v must both be adjacent to another x_i on the graph for $i \neq 1$.

If u and v are both adjacent to x_i , for example, we have a cycle $u \rightarrow x_i \rightarrow x_1 \rightarrow u$ and v is adjacent to two of the vertices on there. (!)

Therefore, u and v must be adjacent to x_i, x_j respectively. WLOG, let $i < j$. Note that, $x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_j \rightarrow v \rightarrow x_1$ is a chord-less cycle where u is adjacent to two of the vertices. (!)

Since we don't have to worry about being color this vertex of degree ≤ 2 , we can remove it from the graph. Repeating this operation, we'd eventually get the graph has no vertices, which is three-colorable.

35 clubs; 7 hours